

Topic: Data Manipulation Language (DML) Commands

Q. Discuss Data Manipulation Language (DML) commands in SQL. Write the syntax and examples for SELECT, INSERT, UPDATE, and DELETE.

📌 Definition to Remember

Data Manipulation Language (DML) is the subset of SQL used to insert, retrieve, modify, and delete the actual **data (records)** residing within database tables. Unlike DDL, DML commands are generally **not auto-committed** — changes can be **rolled back** until a `COMMIT` is issued. (CRUD: **C**reate = INSERT, **R**ead = SELECT, **U**ppdate = UPDATE, **D**elese = DELETE)

DML Commands

1. INSERT

Adds new rows (records) into a table.

```
-- Syntax 1: Specify columns (recommended)
INSERT INTO Student (Roll_No, Name, Grade) VALUES (101, 'Alice', 'A');

-- Syntax 2: All columns in order
INSERT INTO Student VALUES (102, 'Bob', 'B');
```

2. SELECT (Data Query Language — DQL)

Retrieves data from one or more tables.

```
-- Basic syntax
SELECT Name, Grade FROM Student WHERE Grade = 'A';

-- Select all columns
SELECT * FROM Student;
```

3. UPDATE

Modifies existing records in a table.

```
UPDATE Student SET Grade = 'A+' WHERE Roll_No = 101;
```

⚠️ **Warning:** Omitting `WHERE` updates **ALL rows** in the table!

4. DELETE

Removes existing rows from a table but **keeps the table structure**.

```
DELETE FROM Student WHERE Roll_No = 102;
```

⚠️ **Warning:** Omitting `WHERE` deletes **ALL rows** (table structure remains — unlike DROP).

DML vs DDL Comparison

Feature	DML	DDL
Operates On	Data (rows/records)	Structure (tables/schemas)
Auto-Committed	Usually No (can ROLLBACK)	Yes (permanent)
Commands	SELECT, INSERT, UPDATE, DELETE	CREATE, ALTER, DROP, TRUNCATE
Reversible	Yes, with ROLLBACK	No

★ Must-Write Points (for 10 marks)

1. DML manipulates **actual data** (rows/records) inside tables.
2. DML commands are **NOT auto-committed** — changes can be rolled back.
3. **INSERT** : adds new rows; can specify column names (recommended) or use all-column syntax.
4. **SELECT** : retrieves data; use **WHERE** to filter, ***** for all columns.
5. **UPDATE** : modifies existing rows; **MUST** use **WHERE** or all rows will be updated.
6. **DELETE** : removes rows; **MUST** use **WHERE** or all rows will be deleted (structure preserved).
7. CRUD = Create (INSERT), Read (SELECT), Update (UPDATE), Delete (DELETE).

⚡ Quick Recall

DML → Not Auto-committed → INSERT (add rows) → SELECT (read data) → UPDATE (modify rows, use WHERE!) → DELETE (remove rows, use WHERE!) → CRUD